Computer Science & DA

Probability and Statistics

Joint P.M.F and P.D.F

Lecture No. 02

By- Dr. Puneet Sharma Sir

Recap of previous lecture

Topic Joint p.m.f and p. d. f

Topics to be Covered

Topic Marginal and Conditional p.d. f

[NAT]

HW

#Q. Two Dice are rolled simultaneously. Let x denotes the sum of two faces and y the absolute value of their difference. Find joint p.m.f for x and y.

(i) Also evaluate P (x = y)

(ii) P [x = 7 ∩ y ≥ 3]

(iii) P [(x + y = 4) ∩ (x − y = 2)] = ?

(iv) Find marginal distributions of x and y

(v) Find conditional distribution of x given y = 2

P[Sum = 4 & Diff = 2]

= P[3, 1] = $\frac{2}{36}$

= P(x, 1) =

JOINT P.D.F

if x and y are C.R.V then their joint p.d.f is denoted by $f(x,y)$ and we have;

① $f(x,y) \geq 0$; ② $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

③ $P(a \leq x \leq b \cap c \leq y \leq d) = \int_{c}^{d}\int_{a}^{b} f(x,y)\,dx\,dy$

④ if x & y are Ind R.V $\Longleftrightarrow$ then $\boxed{f(x,y) = f(x).f(y)}$

⑤ Marginal density func^n for x is $f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$

⑥ Marginal Density func^n of y is

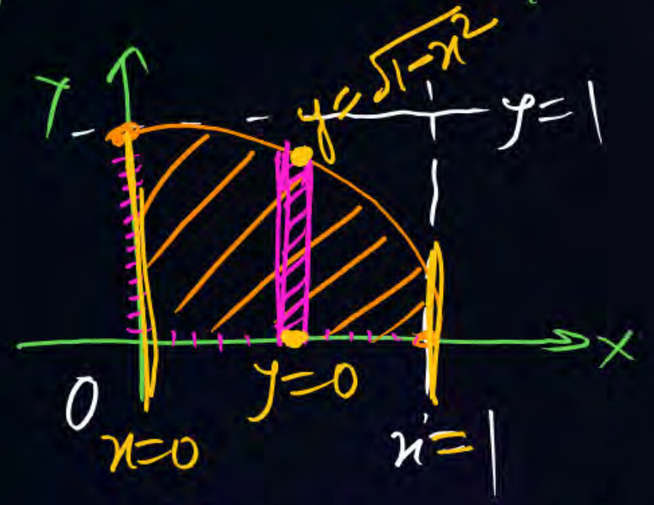
$$f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

⑦ Conditional Density func^n of x for given y is $f(x/y) = \dfrac{f(x,y)}{f(y)}$

⑧ Conditional Density func^n of y for given x is $f(y/x) = \dfrac{f(x,y)}{f(x)}$

#Q. Two point are randomly selected at random in the interval ~~$0 \le x \le 1$~~ $[0,1]$. Find the probability that sum of their squares is less than 1.

Sol: Let $0 \le x \le 1$, $0 \le y \le 1$



Total area $= (1)(1) = 1$

fav case $\Rightarrow x^2+y^2 \le 1$

fav Area $= \frac{1}{4}\pi(1)^2 = \frac{\pi}{4}$

Req Prob $= \frac{f.A}{T.A} = \frac{\pi/4}{1} = \frac{\pi}{4}$

(III-II) $\because x \in [0,1]$ & x is U.R.V then p.d.f, $f(x) = \begin{cases} \frac{1}{1-0}, & 0 \le x \le 1 \\ 0, & \text{otw} \end{cases}$

& $y \in [0,1]$ & y is U.R.V then p.d.f, $f(y) = \begin{cases} \frac{1}{1-0}, & 0 \le y \le 1 \\ 0, & \text{otw} \end{cases}$

$\because$ x and y are also Ind so joint p.d.f

$$f(x,y) = \begin{cases} 1, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otw} \end{cases}$$

$$P[x^2+y^2 \le 1] = \int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}} f(x,y)\,dy\,dx = \int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}} (1)\,dy\,dx$$

$$= \int_{x=0}^{1} \sqrt{1-x^2}\,dx = \cdots\cdots = \frac{\pi}{4}$$

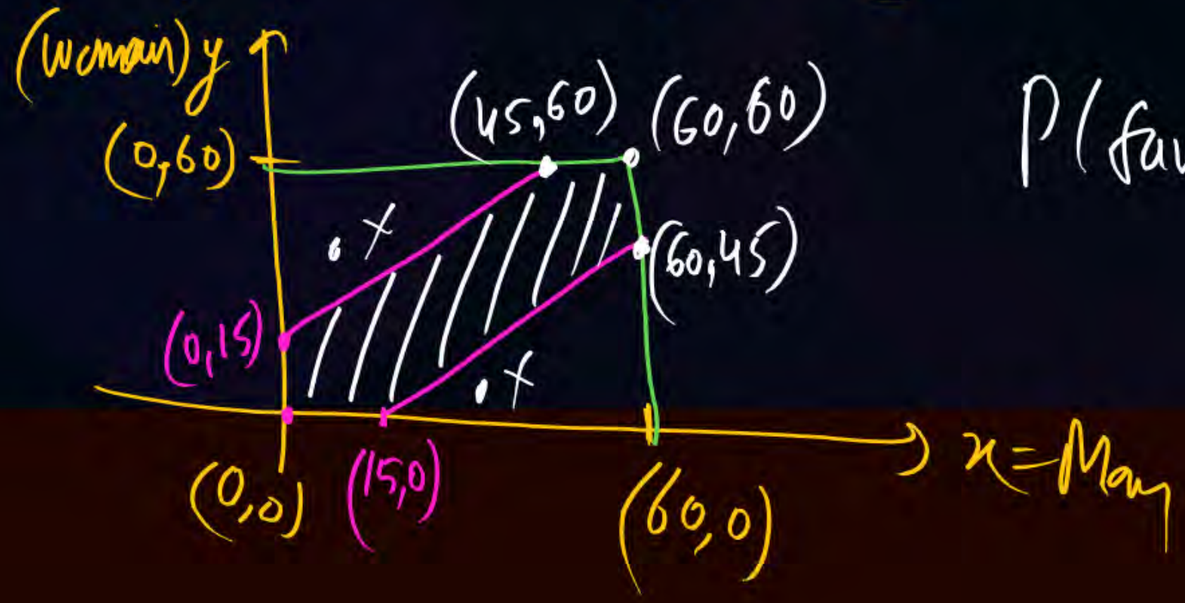
#Q. A couple wants to meet in park between 1 P.M. and 2 P.M. with the understanding that each will wait no longer than 15 minutes for other. What in the probability that they will meet?

(ii) Find the probability that 1st to arrive has to wait longer than 10 minutes?

25/36

M-I $x = \{$ time of arrival of Man $\}$

$y = \{$ " " " of woman $\}$



Total area $= 60 \times 60$

unfav area $= 2$ (area of lower Δ) $= 2\left[\frac{1}{2} \times 45 \times 45\right]$

$= 45 \times 45$

$P(\text{fav area}) = 1 - P(\text{unfav area})$

$= 1 - \left(\frac{45 \times 45}{60 \times 60}\right) = \frac{7}{16}$ Ans

(M-II) $P(X \text{ will not meet } Y) = \frac{45}{60} = \frac{3}{4}$

$P(Y \text{ " " " } X) = \frac{45}{60} = \frac{3}{4}$

$P(\text{they both will not meet}) = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$

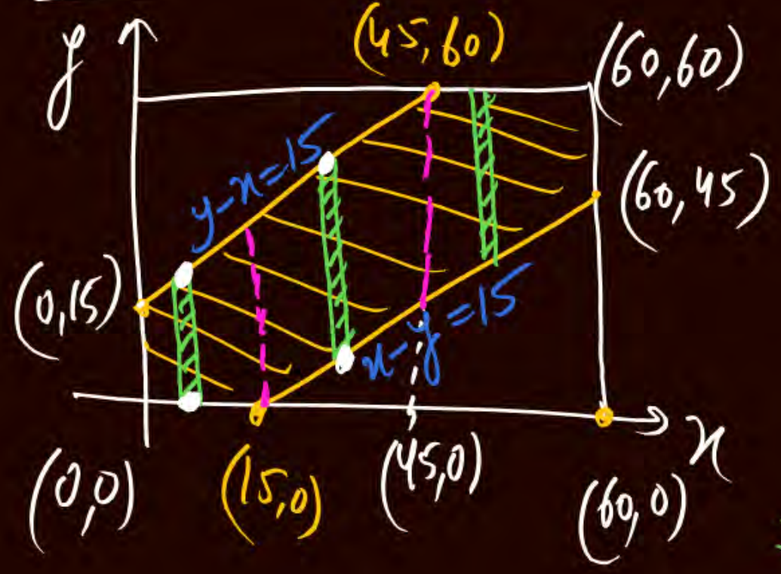
$P(\text{they will meet}) = 1 - \frac{9}{16} = \frac{7}{16}$

(M-III) $x \in (0,60)$ & x is U.R.V $\Rightarrow f(x) = \begin{cases} \frac{1}{60}, & 0<x<60 \\ 0, & \text{otw} \end{cases}$

$y \in (0,60)$ & y is U.R.V $\Rightarrow f(y) = \begin{cases} \frac{1}{60}, & 0<y<60 \\ 0, & \text{otw} \end{cases}$

$\because$ x & y are Ind so joint p.d.f is

$$f(x,y) = f(x).f(y) = \begin{cases} \frac{1}{60 \times 60}, & 0<x<60,\ 0<y<60 \\ 0, & \text{otw} \end{cases}$$



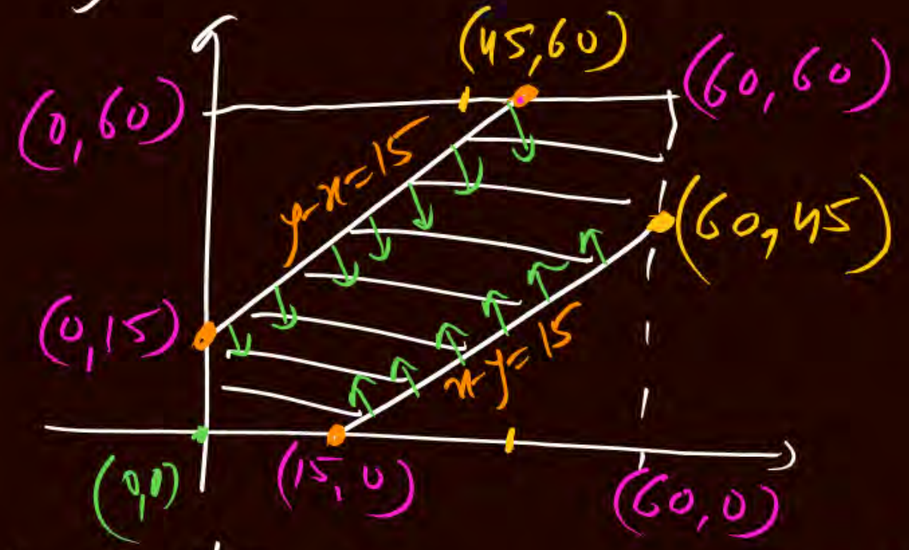
$$P[\text{Shaded Region}] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

$$= \int_{x=0}^{15}\int_{y=0}^{x+15} \left(\frac{1}{3600}\right) dy\,dx + \int_{x=15}^{45}\int_{y=x-15}^{x+15} \left(\frac{1}{3600}\right) dy\,dx + \int_{x=45}^{60}\int_{y=x-15}^{60} \left(\frac{1}{3600}\right) dy\,dx$$

$$= \frac{1}{3600}\left[\int_0^{15}(x+15)\,dx + \int_{15}^{45}(30)\,dx + \int_{45}^{60}(75-x)\,dx\right] = \frac{7}{16}$$

x = time of arrival of M

y = " " " W



(60, 44) ✗

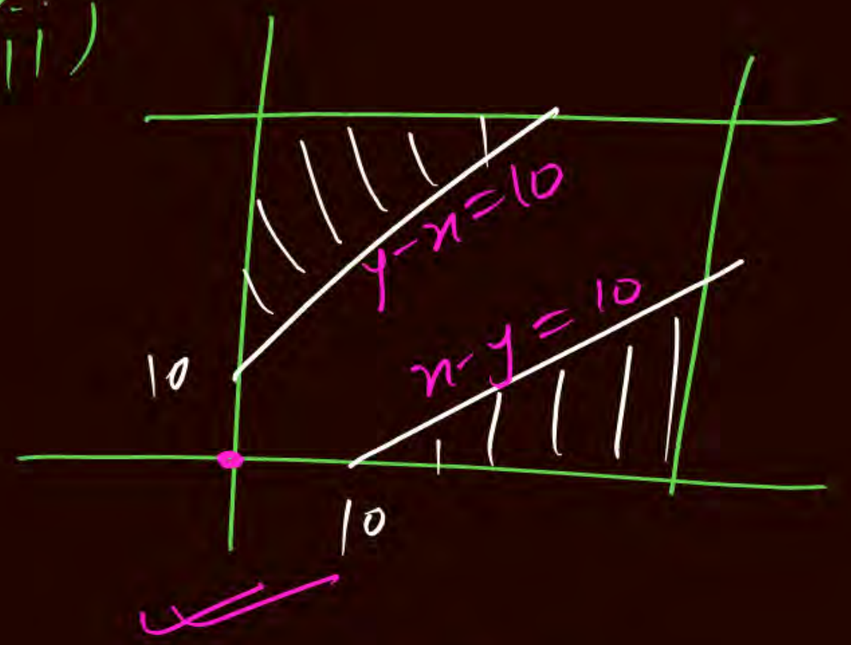
(44 60) ✗

for Meeting $x-y<15 \Rightarrow 0-0<15$ ☺ ✓

or

$y-x<15 \Rightarrow 0-0<15$ ☺ ✓

(ii)



$A_1 = \frac{25}{88}$

#Q. The joint p.d.f of 2D random variable (x, y) is given as:

$$f(x,y)=\begin{cases}2, & 0<x<1, 0<y<x\\ 0, & \text{otherwise}\end{cases}$$

(i) Verify that it is valid p.d.f.

(ii) Find marginal density functions of x and y

(iii) Check for the independence of x and y

(iv) Find the conditional density function of Y for given X = x

(v) Find the conditional density function of X for given Y = y

(iv) $f(y/x)=\frac{f(x,y)}{f(x)}=\frac{2}{2x}=\frac{1}{x}, \quad 0<x<1$

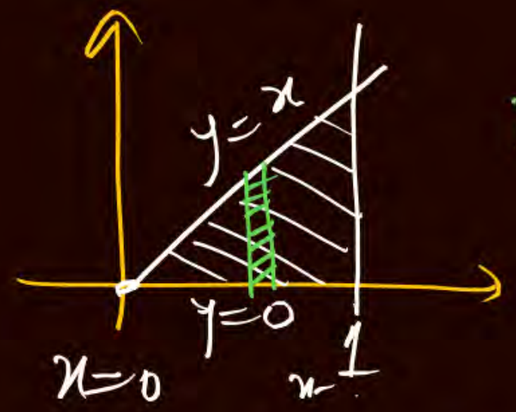
$f(x/y)=\frac{f(x,y)}{f(y)}=\frac{2}{2(1-y)}$

$=\frac{1}{1-y}, \quad 0<y<1$

① $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_{x=0}^{1}\int_{y=0}^{x} (2)\,dy\,dx = 2\int_{x=0}^{1} x\,dx = 2\cdot\left(\frac{x^2}{2}\right)_0^1 = 1$ ☺

② Marginal Density fun^n of x is;

$$f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_{y=0}^{x} 2\cdot dy = 2x$$



i.e. $\boxed{f(x) = 2x\ ;\ 0 \le x \le 1}$

Marginal Density fun^n for y is

$$f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_{x=y}^{1} (2)\,dx$$

$$= 2(x)_y^1 = 2(1-y)$$

i.e. $\boxed{f(y) = 2(1-y)\ ;\ 0 < y < 1}$

③ $\because f(x)\cdot f(y) = (2x)(2(1-y)) = 4x(1-y) \neq f(x,y)$ So Not Ind.

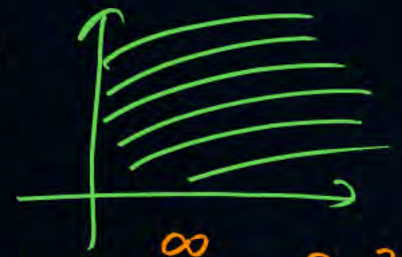
#Q. Joint p.d.f. of x and y is given as; $\boxed{f(x, y) = 4xye^{-(x^2+y^2)}}$, $x \geq 0, y \geq 0$

(i) Verify that it is valid p.d.f.

(ii) Find marginal p.d.f. for x and y

(iii) Check the independency of x and y

(iv) Find conditional density of x for given y.



Note → Evaluate $\int_0^\infty 2x e^{-x^2} dx = ? = \int_0^\infty e^{-t} dt = 1$

① $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_0^\infty\int_0^\infty 4xy e^{-(x^2+y^2)} dx\,dy$

$= \int_0^\infty 2x e^{-x^2} dx \times \int_0^\infty 2y e^{-y^2} dy$

$= 1 \times 1 = 1$

(ii) $f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_0^\infty 4xy e^{-x^2-y^2} dy$

$= 2x e^{-x^2} \int_0^\infty 2y e^{-y^2} dy = 2x e^{-x^2}(1)$

$f(x) = 2x e^{-x^2}$; $0 \leq x < \infty$

Similarly $f(y) = 2y e^{-y^2}$, $0 \leq y < \infty$

(iii) $\because f(x).f(y) = (2x e^{-x^2})(2y e^{-y^2})$

$= 4xy e^{-(x^2+y^2)} = f(x,y)$

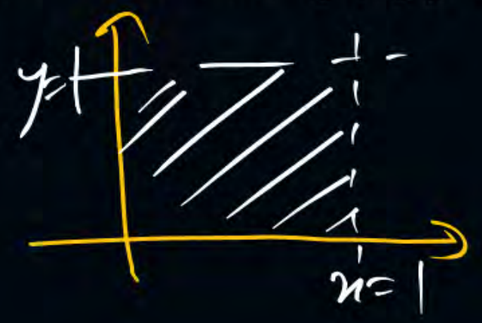
Hence x & y are Ind.

(iv) $f(x/y) = \dfrac{f(x,y)}{f(y)} = \dfrac{4xy\, e^{-(x^2+y^2)}}{2y\, e^{-y^2}} = 2x\, e^{-x^2}, \quad 0 \le x < \infty$

& $f(y/x) = \dfrac{f(x,y)}{f(x)} = \dfrac{4xy\, e^{-(x^2+y^2)}}{2x\, e^{-x^2}} = 2y\, e^{-y^2}, \quad 0 \le y < \infty$

#Q. Suppose that two-D continuous Random variable (x, y) has joint p.d.f given as

$$f(x,y) = \begin{cases} 6x^2y, & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$



(i) Verify that $\int_0^1 \int_0^1 f(x,y)\,dxdy = 1$

(ii) Find $P\left(0 < x < \frac{3}{4}, \frac{1}{3} < y < 2\right)$

(iii) $P(x + y < 1)$

(iv) $P(x > y)$

(v) $P(x < 1/y < 2) = ?$

(vi) Also find their M.D.F and check their independency.

M.D.F for x →

$$f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_{y=0}^{1} 6x^2y\,dy = 6x^2\left(\frac{y^2}{2}\right)_0^1 = 3x^2, \quad 0 \le x \le 1$$

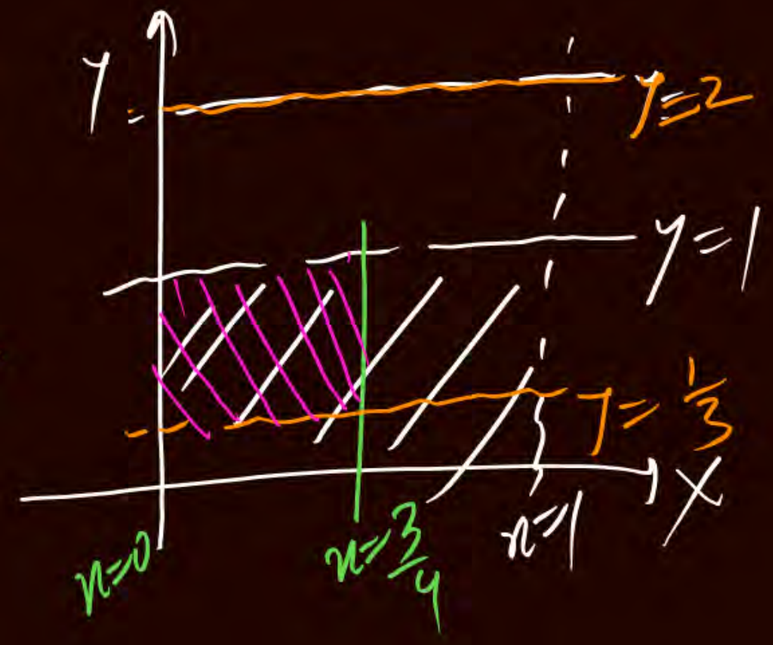
M.D.F for y →

$$f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_{x=0}^{1} 6x^2y\,dx = 6y\left(\frac{x^3}{3}\right)_0^1 = 2y, \quad 0 \le y \le 1$$

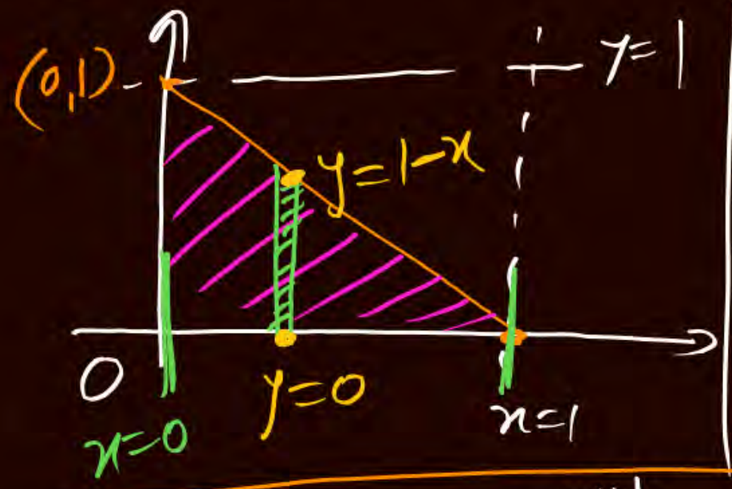
$\because f(x)\cdot f(y) = f(x,y)$ so x & y are Ind

(ii) $P\left[0 < x < \frac{3}{4},\ \frac{1}{3} < y < 2\right] = ? = \int_{y=\frac{1}{3}}^{1}\int_{x=0}^{3/4} (6x^2y)\,dx\,dy$

$$= 6\left[\frac{x^3}{3}\right]_0^{3/4}\left[\frac{y^2}{2}\right]_{1/3}^{1} = \frac{3}{8}$$



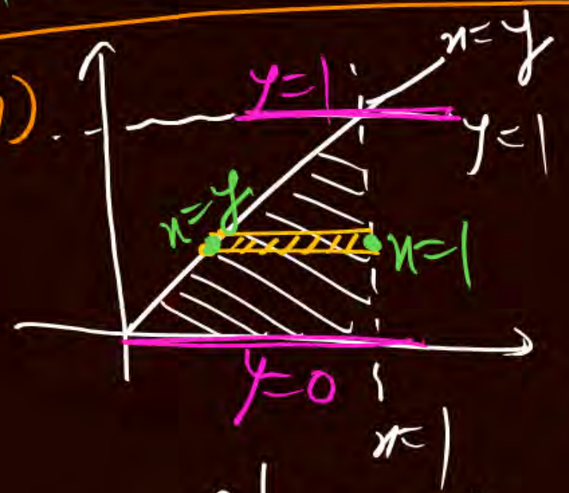
(iii) $P[x+y<1] = ? = \int_{x=0}^{1}\int_{y=0}^{1-x} (6x^2y)\,dy\,dx$



$= \int_{x=0}^{1} 3(1-x)^2 \cdot x^2\,dx$

$= \cdots\cdots = \frac{1}{10}$

(iv)



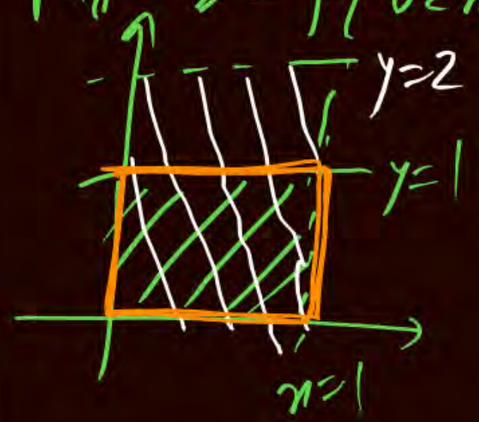
$P[x>y] = ?$

$= \int_{y=0}^{1}\int_{x=y}^{1} 6x^2y\,dx\,dy$

$= \int_{y=0}^{1} 6y\frac{(1-y^3)}{3}\,dy = \cdots = \frac{3}{5}$

(v) $P[x<1/y<2] = \frac{P[x<1 \cap y<2]}{P(y<2)}$ — (1)

$P(y<2) = P[0<x<1, y<2] = \int_0^1\int_0^1 f(x,y)\,dx\,dy$



$= 1$

By (1) $P(x<1/y<2) = \frac{P[0<x<1, 0<y<1]}{P[y<2]}$

$= \frac{1}{1} = 1$

[NAT]

#Q. If Joint p.d.f for x and y is given as $f(x, y) = \begin{cases} \frac{1}{4}(1 + xy), & |x| < 1 \Rightarrow -1 < x < 1 \\ & |y| < 1 \Rightarrow -1 < y < 1 \\ 0, & \text{otherwise} \end{cases}$

Show that x and y are not independent.

M.D.F for 'x' is $f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_{-1}^{1} \frac{1}{4}(1+xy)\,dy = \frac{1}{4}\left[y + x\cdot\frac{y^2}{2}\right]_{-1}^{1} = \frac{1}{2}$

MDF for 'y' is $f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_{-1}^{1} \frac{1}{4}(1+xy)\,dx = \cdots\cdots = \frac{1}{2}$

$\because f(x).f(y) = \frac{1}{4} \neq f(x,y)$ So Not Ind

[NAT]

#Q. If $f(x,y) = \begin{cases} cxy, & 0 \le x \le 2, \text{ and } 0 \le y \le x \\ 0 & \text{otherwise} \end{cases}$ is joint p.d.f. for x and y then evaluate

(i) $P\left(\frac{1}{2} < x < 1\right) = ?$

(ii) $P(y \ge 1) = ?$

(iii) $P\left(\frac{1}{2} < x < 1, y \ge 1\right) = ?$

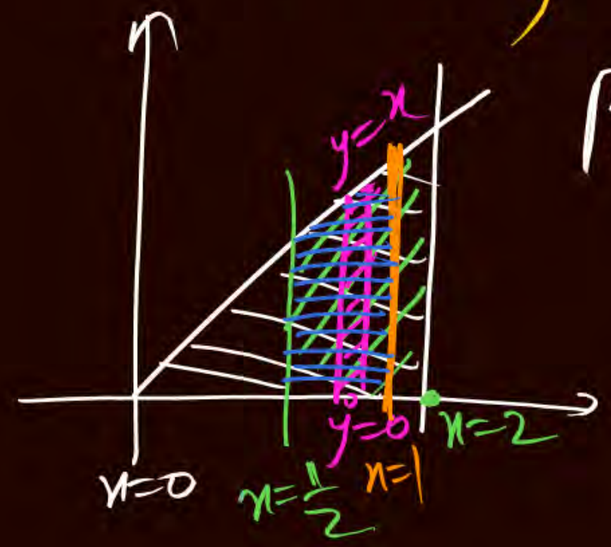
$\because f(x,y)$ is joint p.d.f so $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

$$\int_{x=0}^{2}\int_{y=0}^{x} cxy\,dy\,dx = 1$$

$$\int_{x=0}^{2}\left(\frac{x^2}{2}\right)(cx\,dx) = 1$$

$$\frac{c}{2}\left[\frac{x^4}{4}\right]_0^2 = 1 \Rightarrow \frac{c}{8}[16] = 1 \Rightarrow c = \frac{1}{2}$$

$$f(x,y)=\begin{cases}\frac{1}{2}xy, & 0<x<2,\ 0<y<x\\ 0, & \text{otw}\end{cases}$$



$$P\left\{\tfrac{1}{2}<x<1\right\}=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

$$=\int_{x=\frac{1}{2}}^{1}\int_{y=0}^{x}\left(\frac{1}{2}xy\right)dy\,dx$$

$$=\int_{x=\frac{1}{2}}^{1}\frac{x}{2}\left(\frac{x^2}{2}\right)dx=\frac{1}{4}\left(\frac{x^4}{4}\right)\Big|_{1/2}^{1}=\frac{15}{256}$$

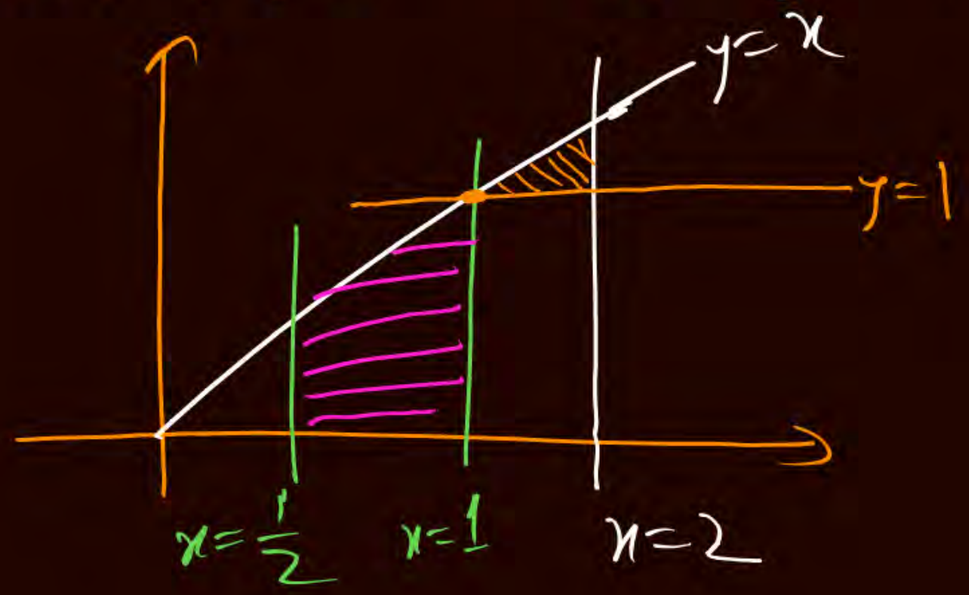
(ii): $P(y\geq 1)=P[1<x<2\cap y\geq 1]$

$$=\int_{x=1}^{2}\int_{y=1}^{x}\left(\frac{1}{2}xy\right)dy\,dx$$

$$=\frac{1}{2}\int_{x=1}^{2}\left(\frac{y^2}{2}\right)_{y=1}^{x}(x\,dx)=\frac{1}{4}\int_{x=1}^{2}(x^2-1)x\,dx$$

$$=\frac{1}{4}\int_{x=1}^{2}(x^3-x)\,dx=\cdots=\frac{9}{16}$$

(iii) $P\left(\frac{1}{2} < x < 1, y \geq 1\right) = ? = 0$ ($\because$ we have no common Region)



[NAT]

#Q. The joint density function of x and y is given as:

$$f(x,y) = \begin{cases} 2e^{-x} \cdot e^{-2y} \; ; 0 < x < \infty, 0 < y < \infty \\ 0 \qquad\qquad\qquad ; \text{otherwise} \end{cases}$$

Find (i) $P(x > 1, y < 1)$

(ii) $P(x < y)$

(iii) $P(x < a)$.

HW

THANK - YOU